A CASE-STUDY BASED REPORT

ON

E-COMMERCE ML INTELLIGENCE PLATFORM: REAL-TIME PURCHASE INTENTION PREDICTION & SALES DEMAND FORECASTING

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Abstract

E-commerce traffic is growing at a rapid pace, yet global digital shopping cart conversion rates consistently remain low, typically hovering between 2.5% and 3.5%. This discrepancy leads to significant revenue losses and inflated customer acquisition costs. Furthermore, merchants face two interconnected, dual-horizon challenges: real-time determination of a visitor's immediate intent to purchase, and long-term macro-level forecasting of inventory demand. Existing analytics solutions rely primarily on post-hoc batch reporting, missing the critical window for real-time customer engagement.

This report details the design, implementation, and evaluation of an end-to-end Machine Learning and Time-Series Analytics system designed to solve both micro-level conversion optimization and macro-level inventory planning. Utilizing a benchmark dataset of 12,330 online shopping sessions from the UCI Machine Learning Repository, the system incorporates rigorous cleaning, duplicate removal, feature engineering, and a cross-validated leakage audit to ensure robust performance. To handle severe class imbalance (15.5% conversion vs. 84.5% non-conversion), gradient boosting models (LightGBM and XGBoost) were trained alongside Logistic Regression baselines using class weighting and precision-recall threshold calibration.

The top-performing classifier, LightGBM (Full Feature Set), achieved an impressive 90.17% accuracy, an ROC-AUC of 0.9344, an F1-score of 69.54%, and a PR-AUC of 0.7476. For cold-start scenarios, a leakage-audited "No PageValues" variant achieved an ROC-AUC of 0.7837. Integrated into a full-stack web application featuring a FastAPI REST API backend and a React 18 frontend dashboard, the platform delivers real-time inference in under 20 milliseconds. The resulting platform bridges academic data science with production-grade enterprise deployment.

Introduction

Background

Modern digital retail operates in a highly competitive environment where acquiring traffic is increasingly expensive. Despite high traffic volumes, cart conversion rates remain remarkably low, averaging between 2.5% and 3.5%. This indicates that over 96% of site visits do not result in a commercial transaction. Traditional web analytics frameworks (such as Google Analytics or

Adobe Analytics) collect massive amounts of clickstream data, but primarily aggregate and report this data in post-hoc, batch-processed reports. Consequently, business insights arrive long after the visitor has departed the site, failing to offer dynamic intervention capabilities while the customer is actively browsing.

Motivation for the Project

Over 70% of e-commerce shopping carts are abandoned prior to final checkout. To mitigate this issue, online merchants frequently issue generic exit-intent popups, indiscriminate discounts, or broad site-wide promotions. However, uncalibrated discounting erodes profit margins and trains shoppers to wait for price reductions. The primary motivation behind this work is to bridge academic data science and full-stack enterprise production by constructing a unified platform. By evaluating micro-level intent (sub-second session classification) and macro-level demand (time-series forecasting), retail platforms can issue tailored incentives—such as free shipping or personalized coupons—only inside a specific visitor's hesitation window, thereby protecting operational margins and lifting overall revenue.

Overview of the Work Carried Out

This study presents an end-to-end Machine Learning intelligence platform. The key engineering phases accomplished include:

- **Data Audit & Cleaning:** Ingested 12,330 clickstream sessions, cleaned records, and eliminated 125 exact duplicates.
- **Feature Engineering & Leakage Audit:** Engineered behavioral interaction metrics and executed a cross-validated leakage audit to ensure predictive validity across early-session and full-session contexts.
- **Model Benchmarking & Calibration:** Trained and calibrated multiple classifier families (LightGBM, XGBoost, Logistic Regression) using custom positive class weighting (scale_pos_weight ≈ 5.4) and Precision-Recall threshold tuning.
- **Forecasting Engine Development:** Implemented lag-feature time-series regressors to project multi-horizon sales demand.
- **Full-Stack Deployment:** Served models via an asynchronous FastAPI REST API backend integrated with a modern React 18 SPA dashboard, delivering inference in under 20 ms.

Scope and Responsible-Use Statement

The system scope covers real-time session inference, batch spreadsheet processing, automated dataset schema detection, and multi-horizon demand plotting. The predictive models operate strictly on anonymized clickstream signals and general session metadata (e.g., operating system, browser, region, page duration). To remain compliant with strict privacy regulations (e.g., GDPR), the platform avoids persistent cross-device tracking or invasive personal identifying information (PII) collection, maintaining ethical data science practices.

Problem Statement

Case-Study Scenario

Online retail operators face severe operational friction due to two distinct, unaligned decision problems:

- **Real-time Visitor Intent (Micro-level):** Answering "Will this current visitor complete a purchase during their ongoing browsing session?" within milliseconds.
- **Macro-level Sales Demand (Supply Chain):** Answering "How much inventory demand is expected over the upcoming daily, weekly, or monthly horizons?" to prevent stockouts or excessive holding costs.

Conventional solutions treat these problems independently using static rule engines or retrospective spreadsheet models, creating delays and inefficiencies.

Central Case-Study Question

"How can machine learning classifiers and time-series forecasters be combined into a unified, low-latency web platform to accurately predict real-time purchase intent and automate sales demand forecasting without introducing feature leakage?"

Formal Problem Definition

Formally, the real-time intent task is modeled as a binary classification problem: given a feature vector X representing clickstream attributes, duration metrics, and session context, estimate the conditional probability $P(Y = 1 \mid X)$, where $Y \in \{0, 1\}$ denotes whether the session culminates in a transaction (Revenue). The demand forecasting task is defined as predicting sales demand Y_{t+h} across forecast horizons $h \in \{1, ..., N\}$ given historical time-series observations Y_t and lagged covariates.

Objectives that Guide the Study

- De-duplicate, audit, and preprocess the 12,330 UCI online shopping session dataset.
- Perform feature engineering and a rigorous 5-fold cross-validated feature leakage audit.
- Address severe class imbalance (84.5% non-purchase vs. 15.5% purchase) using positive class weighting (scale_pos_weight ≈ 5.4) and optimal Precision-Recall threshold calibration.
- Build a time-series forecasting engine with automated schema detection and lag features.
- Deploy the end-to-end framework via FastAPI and React 18 with sub-20 ms latency.

Assumptions and Limitations

The primary classification assumptions rely on benchmark session aggregates from the UCI repository. Known limitations include the absence of raw millisecond-level mouse tracking/hover events, lack of exogenous economic variables (e.g., inflation or competitor pricing in

forecasters), and session-level aggregation constraints enforced by modern cookie privacy standards.

Dataset and Exploratory Data Analysis

About the Dataset

The primary benchmark dataset comprises 12,330 raw online shopping sessions sourced from the UCI Machine Learning Repository. Each record contains 18 initial attributes spanning clickstream engagement, Google Analytics page metrics, and user contextual factors:

- **Page Counts & Durations (Numeric):** Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration.
- **Google Analytics Metrics (Numeric):** BounceRates, ExitRates, PageValues.
- **Temporal & Technical Context:** SpecialDay, Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, Weekend.
- **Target Variable:** Revenue (Boolean indicating completed transaction).

Purpose of EDA

The primary objective of Exploratory Data Analysis (EDA) was to uncover feature distributions, identify correlation structures, detect data duplication, evaluate class imbalance, and isolate potential data leakage vectors.

Preparation Steps Actually Performed

125 exact duplicate sessions were identified and removed during the initial audit, resulting in 12,205 clean records. Missing value validation confirmed zero missing entries. Categorical variables (Month, VisitorType, Weekend, etc.) were processed using OneHotEncoder while numeric signals were scaled via StandardScaler inside a ColumnTransformer pipeline. Engineered interaction features included total_pages, avg_time_per_page, product_page_share, is_peak_season, and has_pagevalue.

EDA Exploration & Leakage Audit

A critical finding from univariate logistic regression and 5-fold cross-validation was the presence of feature leakage associated with PageValues. PageValues reflects average revenue generated by pages visited prior to transaction completion. While highly predictive in late-stage sessions, it is unavailable or zero for cold-start visitors arriving at early funnel stages. To ensure valid real-world deployment, dual model variants were established: "Full" (including PageValues, ROC-AUC 0.9344) and "No_PageValues" (strict cold-start early session mode, ROC-AUC 0.7837).

Summary of EDA Findings

The session distribution exhibits a stark 84.5% (10,422 sessions) non-conversion versus 15.5% (1,908 sessions) conversion split. Addressing this severe ratio required explicit imbalance

correction via positive weight multipliers (scale_pos_weight ≈ 5.4) and calibrated classification cutoffs.

Algorithm Selection and Implementation

Four primary classifier architectures were evaluated across both feature set variants:

- **LightGBM:** Histogram-based gradient boosting using leaf-wise growth; optimized for high speed and categorical efficiency.
- **XGBoost:** Depth-wise decision tree boosting incorporating L1/L2 regularization to resist noisy session features.
- **Logistic Regression:** Linear benchmark trained using class_weight='balanced'.
- **Dummy Classifier:** Naive baseline predicting majority class to establish empirical lower bounds.

Experimental Benchmarks & Results

Model	Variant	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Inference
LightGBM	Full	90.17%	67.49%	71.73%	69.54%	0.9344	0.7476	~14 ms
XGBoost	Full	88.90%	62.70%	71.73%	66.91%	0.9232	0.7183	~13 ms
Logistic Reg.	Full	88.08%	58.83%	79.32%	67.56%	0.9257	0.6918	~8 ms
LightGBM	No PageValues	74.68%	33.84%	64.66%	44.42%	0.7837	0.3722	~12 ms
XGBoost	No PageValues	76.81%	34.46%	53.40%	41.89%	0.7552	0.3335	~16 ms
Logistic Reg.	No PageValues	70.91%	30.66%	68.06%	42.28%	0.7556	0.3327	~7 ms
Dummy Baseline	Both	15.65%	15.65%	100.0%	27.06%	0.5000	0.1565	<1 ms

System Architecture & Full-Stack Implementation

The platform was engineered as a decoupled microservice framework:

- **Frontend Presentation Layer:** Built with React 18, Vite, Tailwind CSS, Recharts for interactive forecast visualizations, and Lucide Icons.
- **Backend Application Layer:** FastAPI async Python service hosting lightweight endpoints (/api/purchase/predict, /api/datasets/upload, /api/forecast/run) enforcing strict Pydantic contract validation.

- **Model Pipeline & Storage:** Pre-compiled Joblib pipeline registries maintaining model artifacts for real-time inference under 20 ms.

Conclusion

This case study successfully designed and deployed a dual-engine Machine Learning platform addressing real-time purchase intention prediction and automated multi-horizon demand forecasting. By addressing class imbalance through custom positive weighting (scale_pos_weight ≈ 5.4) and eliminating feature leakage between full and cold-start variants, the top LightGBM classifier achieved 90.17% accuracy, 0.9344 ROC-AUC, and sub-20 ms inference latency. Connected to a React 18 dashboard and a FastAPI backend, the solution provides an efficient, scalable, and low-cost mechanism for conversion optimization and intelligent inventory management.

Signature of the Team members